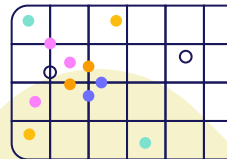
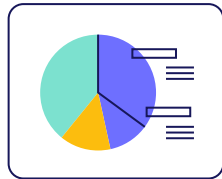
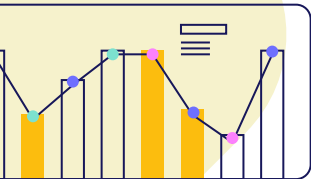


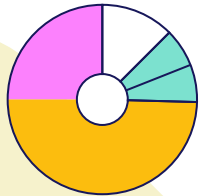
Demand Estimation & Market Analysis: Air Fryers

Joelle Zamperini (jxj5jz), Diya Patel(asq7mf), Elizabeth Albarran (ezu7cg), Brishti Chakraborty (jmc3kn), Chayun Kim (xqj2mn)

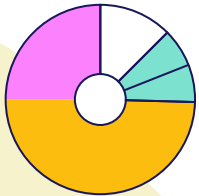
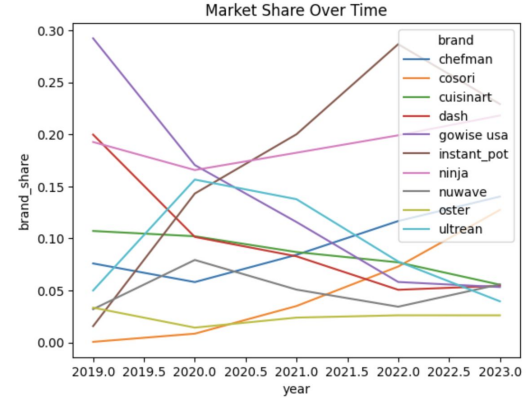
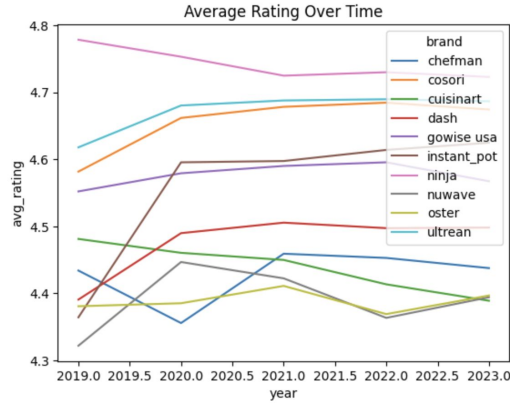
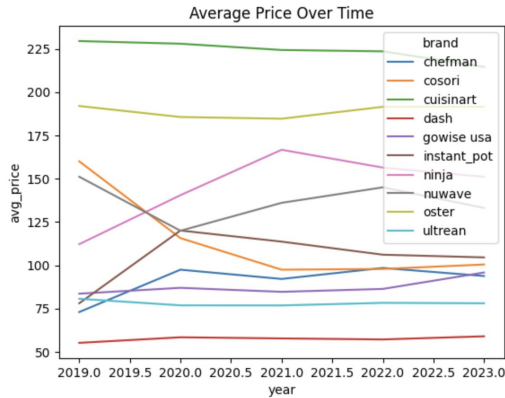


Short Description of the Air-Fryer Market

- 2019-2023: concentrated consumer appliance market dominated by a small set of leading brands
- Moderate stability
- Some brands maintain high market share
- Prices Vary significantly across brands → segmentation between premium and budget options
- Overall high consumer ratings
- Competition driven by price, features, product characteristics, and brand positioning.

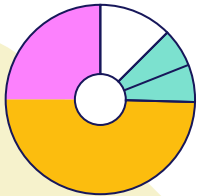


Prices, Ratings, and Market Shares Over Time



The Demand Model

- We estimate demand using a multinomial logit model, implemented as a linear regression on log market share
 - Dependent variable: log of brand market share
 - Independent variables: price, rating, product characteristics
 - Controls: brand fixed effects and year fixed effects
- Interpretation:
 - Higher utility $\rightarrow$ higher market share
 - Price lowers utility
 - Better features and ratings increase utility
- This model allows us to quantify how sensitive customers are to price and product attributes.



Estimated Price Coefficient

Our estimated price coefficient is **-0.0377**

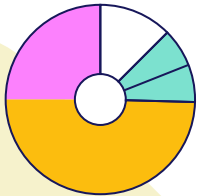
This means that higher prices tend to lead to lower market share

Why this matters:

- The negative sign reflects a downward-sloping demand
- Higher price → lower demand

Strategic Importance:

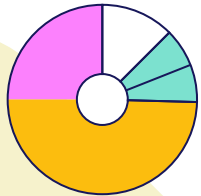
- Firms must balance pricing carefully, as even a small price increase reduces demand and impacts profitability.



Most Important Product Features

The product feature results suggest that demand is strongest for air fryers with broad, practical appeal. The two strongest positive features are window share and compact share, meaning brands with more window-style and compact products tend to have higher market shares after controlling for price, rating, brand, and year. Oven-style products are also positively associated with demand, but the effect is smaller.

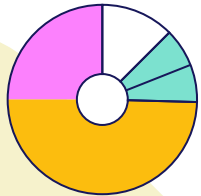
By contrast, dual-basket and rotisserie features have negative coefficients. This does not necessarily mean these features are bad, but they appear to be more niche and may not appeal to the average buyer in the same way that compact, easy-to-use designs do. From a business perspective, the strongest market opportunity seems to be in products that are convenient, compact, and easy for consumers to understand.



Estimated Unit Costs, Markups, and Share-Weighted Profits

Using the estimated price coefficient of -0.0377 , we backed out demand slopes, unit costs, markups, and share-weighted profits for each brand-year. The negative price coefficient is important because it means demand falls as price rises, which makes the markup and cost estimates economically meaningful.

The markup is calculated as the difference between price and estimated unit cost, while share weighted profit multiplies that markup by the brand's market share. This means brands with both strong margins and high market shares perform best in the profit index. Premium brands, such as Cuisinart, likely have higher estimated unit costs because their average prices are much higher, while value or volume-oriented brands like Dash and GoWISE USA may rely more on large market shares to generate profit.



Conclusion: Brands' Relative Market Positions

Overall, the air-fryer market appears to have a mix of premium-positioned brands, volume/value brands, and feature specialized brands. Cuisinart has the strongest brand fixed effect in the demand model, followed by Ninja and Instant Pot, suggesting these brands have strong brand power even after accounting for price, ratings, and product characteristics.

Dash and GoWISE USA appear to be more volume oriented, with strong market shares and lower to mid price positioning. Chefman stands out as more feature diverse, especially in rotisserie and window-style products, but those niche features do not necessarily translate into the strongest demand. Overall, Cuisinart and Ninja look strongest from a brand-positioning perspective, while Dash and GoWISE USA look important from a market-share and volume perspective.

